Simulations for Harnessing the VO_2 Phase Transition for Automatic Gain Control in Transimpedance Amplifiers

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Contents

1 Yuanhang-based current-drive validation

This run checks that the Yuanhang-based implementation can produce a stable current-driven oscillation with a nonzero voltage valley. It validates the simulation mechanism; it is not a fit to the experimental sample.

All values were taken from Zhang *et al.* [1], except C_{th}. At 600 μA, the nominal value (49.6278 pJ/K) did not sustain oscillations, so it was increased fourfold to slow the thermal response and obtain a stable validation cycle. This control value is not an estimate of our sample's thermal capacitance.

Quantity	Value
Current pulse	600 μA from 2 to 38 μs
Simulation window	0–40 μs
Time step	0.5 ns
Initial state	$V(0) = 0$ V, $T(0) = 324.9$ K, insulating branch
Electrical capacitance	$C = 145.34619293$ pF
Thermal capacitance	$C_{\text{th}} = 198.51107324$ pJ/K (4× control)
Environmental conductance	$S_e = 0.20558726$ mW/K
Ambient temperature	$T_0 = 325$ K
Metallic resistance	$R_m = 1286.3165\ \Omega$
Noise and thermal neighbor coupling	zero
Analysis interval	10–38 μs

Table 1: Parameters used for the validation run. All unlisted resistance and hysteresis parameters retain their Yuanhang values.

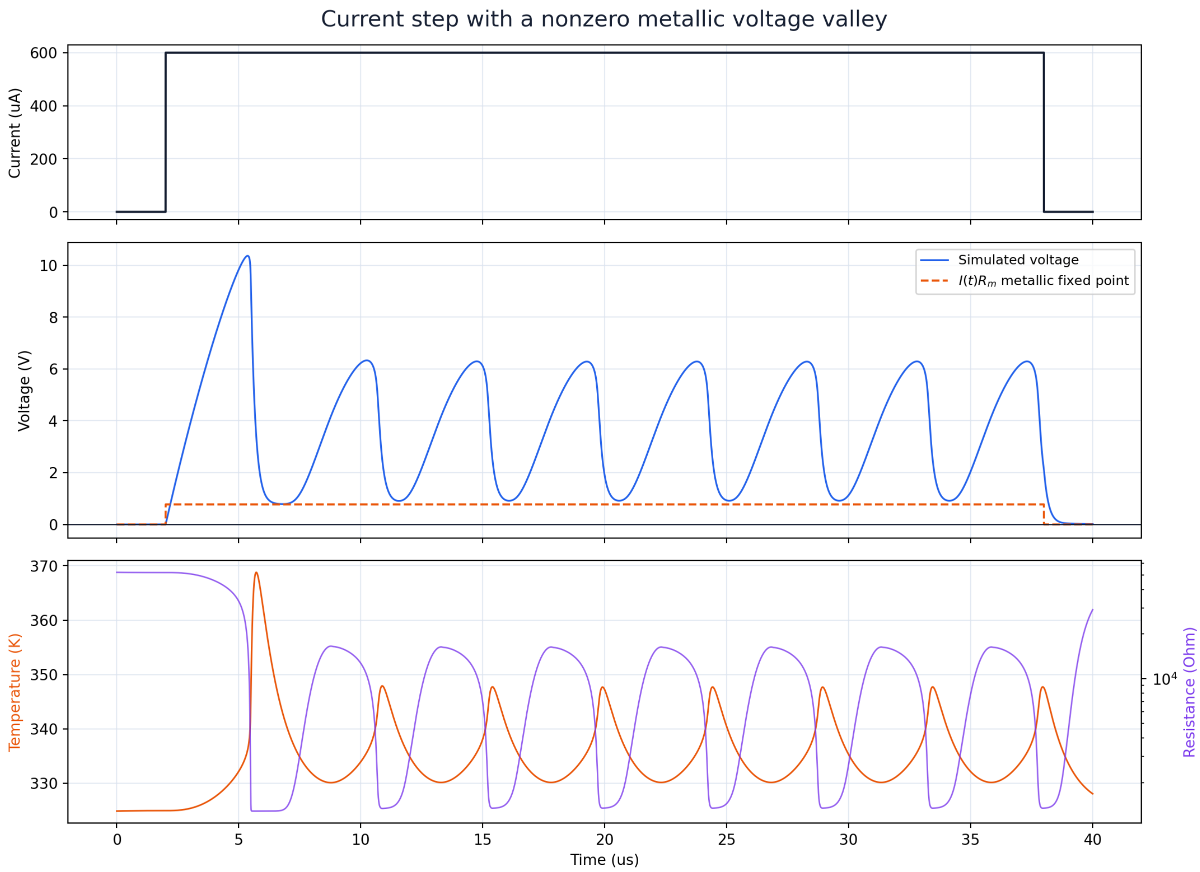

Figure 1: Applied current and simulated voltage, temperature, and resistance. The dashed orange line is the instantaneous metallic fixed point $I(t)R_m$. The analysis excludes the initial charging transient and uses the stable interval from 10 to 38 μs.

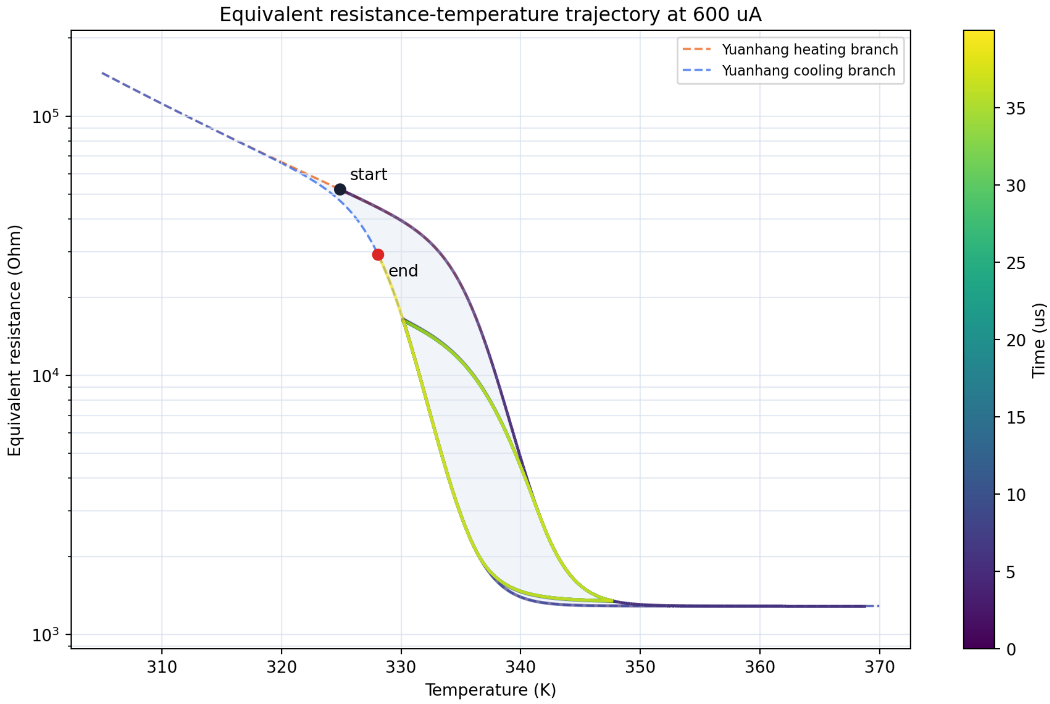

Figure 2: Equivalent resistance versus temperature. The dashed curves and shaded band show Yuanhang's major heating and cooling hysteresis; color shows progression through the 40 μs simulated trajectory.

Metric	Result
Oscillation detected	yes
Detected cycles	5
Frequency	0.221685 MHz
Period	4.51090 μs
Period coefficient of variation	1.77×10^{-4}
Steady voltage range	0.905654–6.33281 V
Full-pulse temperature range	324.9–368.8 K
Steady temperature range	330.16–347.91 K
Steady resistance range	1.339–16.382 kΩ
Metallic electrical time CR_m	186.961 ns
Thermal time C_{th}/S_e	0.965581 μs

Table 2: Metrics measured from the validation run.

The metallic-state voltage is set approximately by $V = IR_m$:

$$V_{\text{metal}} \approx IR_m = (600 \times 10^{-6}\ \text{A})(1286.3165\ \Omega) = 0.771790\ \text{V}. \tag{1}$$

The simulated minimum is 0.905654 V; it remains above this fixed point because switching reverses before the capacitor settles. The experimental manuscript reports 40–60 MHz oscillations [2], whereas this validation gives 0.221685 MHz. Sample-specific resistance and time constants are therefore still required for a quantitative match.

2 Sample-specific resistance–temperature fit

The supplied file contains 264 measurements from one cooling sweep and one heating sweep over 289.85–370.16 K at 2 μA. We fitted the major-loop form

$$R(T,\delta) = R_m + R_0 e^{(E_a/k_B)/T}\left[\frac{1}{2}+\frac{1}{2}\tanh\Big(\beta\left(\delta\frac{w}{2}+T_c-T\right)\Big)\right], \qquad \delta=\begin{cases}-1, & \text{cooling,}\\ +1, & \text{heating.}\end{cases} \tag{2}$$

The fit minimizes the residual in $\log_{10} R$. The minor-loop parameter γ was fixed to Yuanhang's value because a major loop alone does not determine it.

Here R_0 is the Arrhenius prefactor, not the insulating resistance. On the fully insulating branch,

$$R_{\text{ins}}(T) \approx R_m + R_0 e^{(E_a/k_B)/T}. \tag{3}$$

For example, the fitted sample gives $R_{\text{ins}}(300\text{ K}) \approx 3.78$ kΩ, well above the metallic floor $R_m = 18.22\ \Omega$.

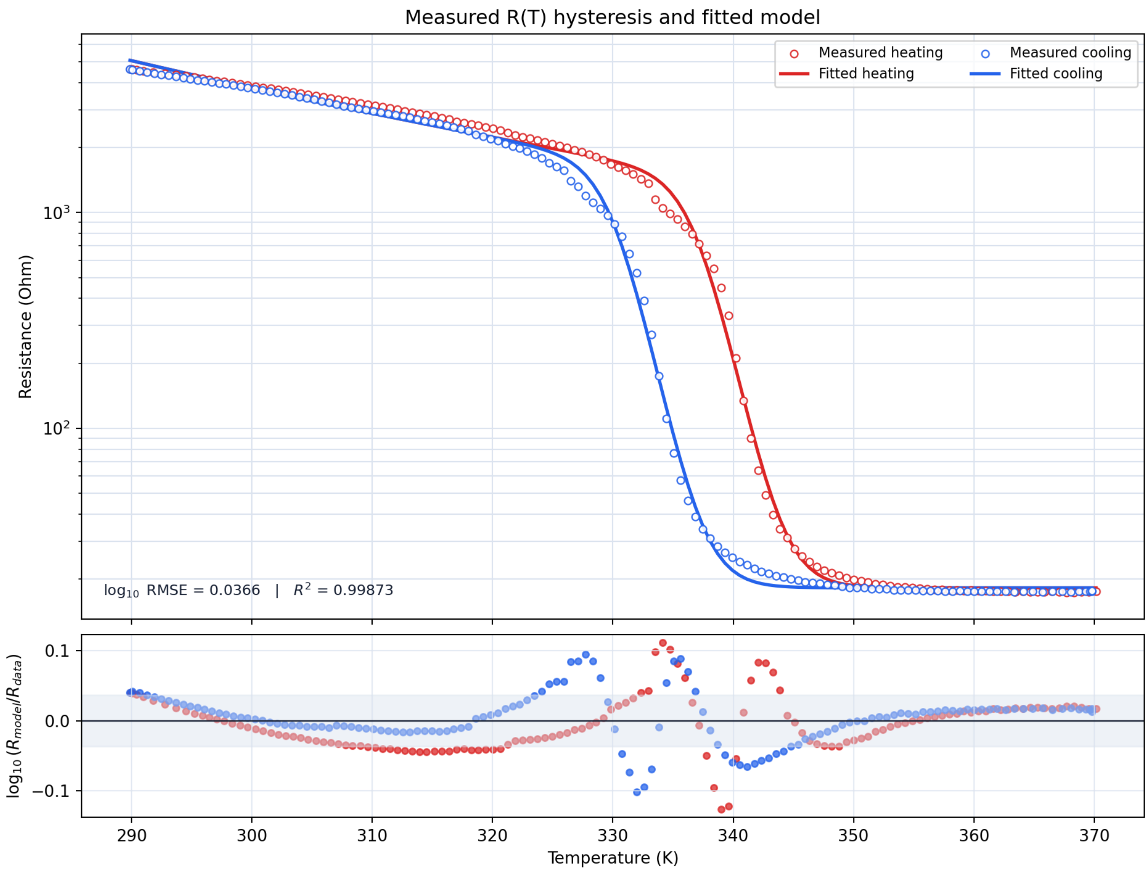

Figure 3: Measured heating and cooling branches, fitted model, and log-resistance residuals.

Parameter	Estimate	95% interval	Yuanhang
R_0 (prefactor)	0.7989 Ω	0.3774–1.8483 Ω	0.005359 Ω
E_a/k_B	2536.97 K	2272.55–2767.52 K	5220.47 K
E_a	0.21862 eV	0.19583–0.23849 eV	0.44987 eV
R_m (metallic floor)	18.2153 Ω	17.4943–18.9010 Ω	1286.32 Ω
T_c	333.494 K	333.137–333.838 K	332.806 K
w	6.8823 K	6.6291–7.2681 K	7.1936 K
β	0.29916 K^{-1}	0.28092–0.32021 K^{-1}	0.25280 K^{-1}
$T_{\text{cool}} = T_c - w/2$	330.052 K	329.642–330.385 K	329.209 K
$T_{\text{heat}} = T_c + w/2$	336.935 K	336.534–337.348 K	336.403 K
γ	0.95627 (fixed)	—	0.95627

Table 3: Sample-specific resistance parameters compared with Yuanhang's reference implementation, which uses the effective metallic resistance $R_m = 1286.32$ Ω. R_0 and E_a/k_B are strongly correlated; their joint combination determines the insulating branch.

The fit gives $R^2 = 0.998734$ and a $\log_{10} R$ RMSE of 0.03663, equivalent to a typical multiplicative error of 1.088. The largest deviation is a factor of 1.336 inside the switching region; this residual is retained rather than adding parameters unsupported by the measurement.

3 Experimental timing and ambient temperature

The sample-specific simulations will reproduce the measured pulse timing rather than use an instantaneous current step. The TIA sweep is taken to share the ambient range reported for the related measurements [2].

Input	Value
Ambient temperature	$T_0 \approx 314.4$ K (314.25–314.55 K)
Pulse width	300 ns
Current rise time	26–27 ns (10–90%)
Current fall time	18–23 ns (90–10%)

Table 4: Measured environmental and timing inputs retained for later simulation of the experiment.

4 Environmental thermal conductance

The objective is to estimate the environmental thermal conductance S_e, which sets the heat-loss rate $S_e(T - T_0)$. The estimate requires a hot operating point whose temperature is approximately stationary. We use the following procedure:

1. Order the measured records by their baseline-corrected input current and identify the last record without coherent oscillation. This gives the stable point closest to switching, where the temperature rise is large but a quasi-steady balance remains defensible.
2. Subtract the pre-pulse current and voltage offsets and select a settled window in which V/I is nearly constant. In this window $dV/dt \approx 0$, so the measured input current is approximately the resistive current through the VO_2 device.

3. Calculate the effective resistance $R_* = V_*/I_*$ and Joule power $P_* = I_*V_*$. Invert the fitted heating branch, $R_{\text{heat}}(T_*) = R_*$, to infer the device temperature T_*.
4. Apply the quasi-steady thermal balance $C_{\text{th}}\, dT/dt = P_* - S_e(T_* - T_0) \approx 0$, which gives $S_e = P_*/(T_* - T_0)$.

The filenames identify the nominal source-voltage setting, not the voltage across the VO_2 device and not the simulation input. We therefore retain that setting only for traceability and use the measured current. Over the common 50–250 ns analysis window, the 250 and 300 mV source settings produced baseline-corrected current steps of 189.6 and 228.2 μA, respectively.

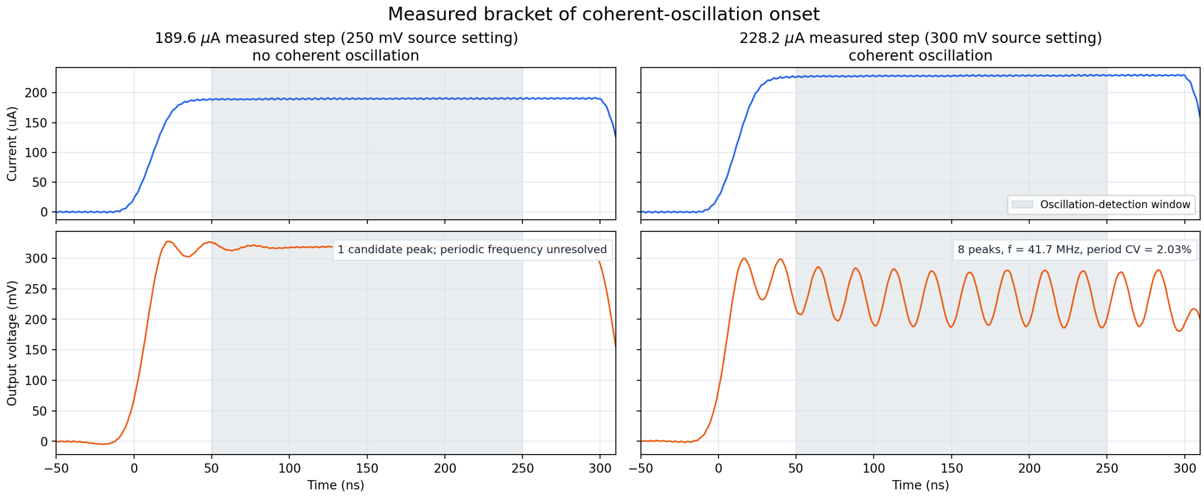

Figure 4: Adjacent baseline-corrected records bracketing oscillation onset on shared axes. The 189.6 μA measured current step (250 mV source setting) contains only one candidate peak, so no periodic frequency is assigned. At 228.2 μA (300 mV source setting), eight peaks give 41.7 MHz with a 2.03% period coefficient of variation.

Figure 4 implements the first step of the procedure. The 228.2 μA record is the first trace with coherent oscillation, while the immediately preceding 189.6 μA record does not oscillate. We therefore selected the latter as the closest measured point below switching onset. We did not use the first transition during the current edge because the device is still heating there and $dT/dt = 0$ would not be justified.

For the selected trace, the current and voltage offsets are their medians from -200 to -50 ns. We then use the settled 100–250 ns interval. Its effective resistance changes by only 0.145%, supporting the quasi-steady approximation. Figure 5 shows the waveform, the stability test, and the temperature inversion.

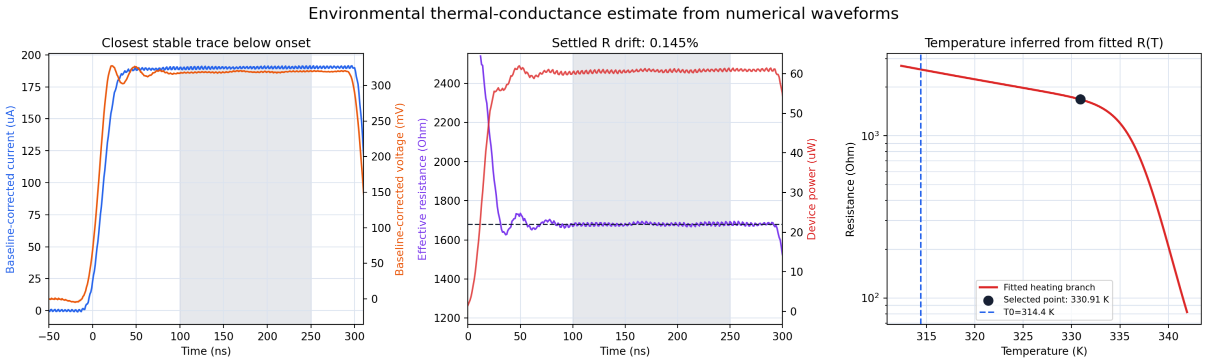

Figure 5: Environmental-conductance evidence. Gray shading marks the settled interval. The measured $R_* = V_*/I_*$ is mapped to T_* through the fitted heating branch.

The baseline-corrected medians give

$$I_* = 190.162\ \mu\text{A}, \qquad V_* = 319.213\ \text{mV}, \qquad R_* = \frac{V_*}{I_*} = 1679.759\ \Omega, \tag{4}$$

and

$$P_* = I_* V_* = 60.658\ \mu\text{W}. \tag{5}$$

Solving the fitted heating branch $R_{\text{heat}}(T_*) = R_*$ gives $T_* = 330.905$ K. At a quasi-steady point, the thermal equation reduces to

$$C_{\text{th}} \frac{dT}{dt} = P_* - S_e(T_* - T_0) \approx 0, \qquad S_e = \frac{P_*}{T_* - T_0}. \tag{6}$$

Therefore,

$$S_e = \frac{60.658\ \mu\text{W}}{330.905\ \text{K} - 314.4\ \text{K}} = 3.675\ \mu\text{W/K} = 0.003675\ \text{mW/K}. \tag{7}$$

Result	Value
Selected trace	190.162 μA settled current (250 mV setting)
First oscillating trace	228.2 μA current step (300 mV setting)
Settled resistance drift	0.145%
Inferred temperature	330.905 K
Temperature rise	16.505 K
S_e estimate	0.003675 mW/K
Conditional 95% interval	0.003434–0.004085 mW/K
Yuanhang reference	0.205587 mW/K

Table 5: Environmental thermal-conductance calculation and comparison.

The interval propagates paired waveform block resampling, the 1000-sample $R(T)$ fit bootstrap, and the 314.25–314.55 K ambient range. Omitting baseline correction gives 0.003560 mW/K, inside the interval. The result remains conditional on the $R(T)$ curve and TIA traces belonging to the same device and on the quasi-static heating branch applying under electrical drive. Yuanhang's conductance is about 56 times larger.

5 Electrical capacitance

The objective is to estimate the effective electrical capacitance C, which stores charge at the device node and sets the electrical response time $\tau_{el} = R(T)C$. In the ideal-current model, the measured input current splits between the VO_2 resistance and this capacitance:

$$I_{in} = \frac{V}{R(T)} + C\frac{dV}{dt}. \tag{8}$$

We therefore seek the part of the measured current that cannot be explained by the instantaneous resistive current $V/R(T)$. The analysis uses low-current, non-switching rising edges so that the device begins on the known cold heating branch and a phase transition does not complicate the electrical transient. It assumes that the current and voltage channels are time-aligned, their pre-pulse offsets are removable, C is approximately constant over the fit window, and the reduced circuit contains the relevant current paths. The procedure is:

1. Subtract each trace's pre-pulse current and voltage medians and select the initial rising edge of the 100–250 mV source-setting records, all of which remain below switching.
2. Use $C_{shortcut} = \Delta I/(dV/dt)$ as a preliminary scale estimate. This expression is valid only if the resistive current is negligible; it must not be accepted before checking the full current balance.
3. Evaluate the physically complete residual $I_C = I_{in} - V/R(T)$ using the fitted cold resistance, and fit $I_C = C\, dV/dt$ over the early edge with the physical constraint $C \geq 0$.
4. Check the result independently against the measured current–voltage timing. If no positive lag is resolved, convert the timing resolution Δt into an upper bound $C \lesssim \Delta t/R(T_0)$. We adopt that bound as a conservative modeling value while retaining $C = 0$ as the best-fit lower endpoint.

We first reproduce the preliminary shortcut. For each trace, ΔI is the baseline-to-plateau current step and dV/dt is a linear fit over 0–30 ns, giving

$$C_{shortcut} = \frac{\Delta I}{dV/dt}, \qquad \frac{\mu A}{mV/ns} = pF. \tag{9}$$

Drive setting	ΔI (μA)	dV/dt (mV/ns)	$C_{shortcut}$ (pF)
100 mV	76.171	3.562	21.382
150 mV	114.172	6.127	18.634
200 mV	152.162	8.235	18.477
250 mV	189.601	9.008	21.047
Median			19.841

Table 6: Reconstruction of the original 19.8 pF shortcut estimate.

The median shortcut estimate is 19.8 pF, but it is biased because the measured current itself rises over 26–27 ns and most of it flows resistively. Applying the full current balance gives

$$C\frac{dV}{dt} = I_{in} - \frac{V}{R(T)}, \qquad C = \frac{I_{in} - V/R(T)}{dV/dt}. \tag{10}$$

At $T_0 = 314.4$ K, the fitted heating branch gives $R(T_0) = 2.570$ kΩ. Substituting the measured current and voltage leaves no resolved capacitive current: the unconstrained early-edge fit is approximately -0.25 to -0.10 pF, depending on the fit window, and the physical nonnegative optimum is therefore $C = 0$. The current and voltage 50% crossings differ by only -0.74 to $+0.52$ ns. Taking 1 ns as the timing resolution gives the non-statistical bound

$$C \lesssim \frac{1\ \text{ns}}{2.570\ \text{k}\Omega} \approx 0.39\ \text{pF}. \tag{11}$$

In contrast, 19.8 pF would give $R(T_0)C = 50.9$ ns and a large voltage lag that is absent from the data. A larger capacitance would worsen the disagreement. This does not establish that the physical capacitance is exactly zero; it means that a positive value is not identifiable from the present traces. To avoid assuming an instantaneous electrical response, the sample-specific simulations and downstream thermal fit use the conservative upper-bound value

$$C_{\text{adopt}} = 0.39\ \text{pF}, \qquad I_R = I_{\text{in}} - C_{\text{adopt}}\frac{dV}{dt}. \tag{12}$$

This is not a measured central estimate: $C = 0$ remains the constrained best fit, and 0.39 pF intentionally maximizes the electrical memory still compatible with the approximately 1 ns timing resolution.

This conclusion is independent of the present choice of γ. The capacitance fit uses the initial cold heating edge, before a temperature reversal; γ modifies only minor hysteresis loops after a reversal and therefore does not enter this calculation. Changing γ can alter later oscillation shapes and frequencies, but it cannot supply the missing early-edge voltage lag. A numerical nonzero estimate requires a faster, low-amplitude nonswitching edge with calibrated current–voltage channel timing; with R approximately constant, $C = \tau_{\text{el}}/R$.

6 Thermal time constant and thermal capacitance

The objective is to estimate the effective thermal capacitance C_{th}, which relates net heating power to the device's temperature rate of change. Together with the environmental conductance found in Section 4, it sets the thermal response time

$$\tau_{\text{th}} = \frac{C_{\text{th}}}{S_e}. \tag{13}$$

A larger C_{th} means that more energy is required to change the device temperature and therefore produces a slower thermal response. Section 5 found no resolved positive electrical capacitance but established the upper bound $C_{\text{adopt}} = 0.39$ pF. We use this conservative value and first separate the resistive current from the capacitive current. The fitted heating branch then acts as a thermometer:

$$I_R(t) = I_*(t) - C_{\text{adopt}}\frac{dV_*}{dt}, \qquad R_*(t) = \frac{V_*(t)}{I_R(t)}, \qquad T_*(t) = R_{\text{heat}}^{-1}(R_*(t))\,. \tag{14}$$

The estimate assumes a single lumped device temperature, constant C_{th} and S_e over the fitted interval, the same sample for the electrical and $R(T)$ measurements, and validity of the fitted heating branch before any thermal reversal. It is conditional on the adopted 0.39 pF upper bound and requires baseline-corrected, time-aligned current and voltage channels. The procedure is:

1. Select moderate non-switching traces with sufficient signal, subtract their pre-pulse offsets, and smooth the oscilloscope noise before evaluating dV_*/dt and forming V_*/I_R. We use measured current steps of 76.2, 114.2, and 152.2 μA (the 100, 150, and 200 mV source settings); the 189.6 μA record (250 mV setting) is reserved as a near-transition sensitivity check.

2. Compute $I_R = I_* - C_{\text{adopt}} dV_*/dt$, reconstruct the measured temperature trajectory from $R_*(t) = V_*(t)/I_R(t)$ and the inverse fitted heating branch, and calculate the applied Joule power $P_*(t) = I_R(t)V_*(t)$.

3. For trial values of C_{th}, integrate the thermal balance using the previously estimated S_e and T_0. Fit one shared C_{th} by minimizing the temperature mismatch across all three primary traces over 15–35 ns, then obtain $\tau_{\text{th}} = C_{\text{th}}/S_e$.

4. Test robustness by fitting each trace separately, checking the excluded 189.6 μA record, varying the fit window, and propagating the $R(T)$, S_e, ambient temperature, and waveform uncertainties.

The thermal trajectory for each trial value is obtained from

$$C_{\text{th}} \frac{dT}{dt} = P_*(t) - S_e(T - T_0), \qquad C_{\text{th}} = S_e \tau_{\text{th}}. \tag{15}$$

A nine-sample, second-order Savitzky–Golay filter is used in the reconstruction. Because the fit ends before a thermal reversal, γ does not enter this estimate. Figure 6 compares the temperatures inferred directly from the electrical measurements with the trajectories predicted by the shared thermal model.

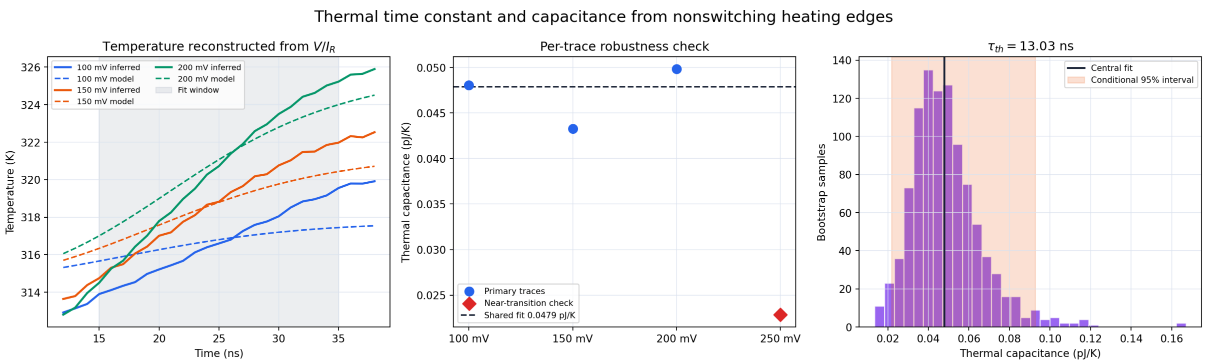

Figure 6: Thermal-capacitance evidence conditioned on $C_{\text{adopt}} = 0.39$ pF. Solid curves are temperatures inferred from V_*/I_R; dashed curves are the shared thermal model. The 189.6 μA measured current step (250 mV source setting) is retained only as a near-transition sensitivity check.

Quantity	Result
Ambient temperature	314.4 K
Environmental conductance	0.00367513 mW/K
Adopted electrical capacitance	0.39 pF (upper bound)
Selected traces	76.2, 114.2, and 152.2 μA steps
Fit interval	15–35 ns
Thermal time constant τ_{th}	13.026 ns
Conditional 95% interval for τ_{th}	5.865–25.174 ns
Thermal capacitance C_{th}	0.047873 pJ/K
Conditional 95% interval for C_{th}	0.021918–0.092624 pJ/K
Temperature-fit RMSE	1.145 K
Individual selected-trace range	0.043261–0.049830 pJ/K
189.6 μA sensitivity result	0.022858 pJ/K; RMSE 2.380 K
Measured oscillation period	16–24 ns
Yuanhang reference	49.6278 pJ/K

Table 7: Thermal time-constant and capacitance results.

The conditional interval propagates trace resampling, the $R(T)$ bootstrap, the conductance bootstrap, the ambient-temperature range, and ± 2 ns changes to the fit window. The archived conductance samples do not retain their paired $R(T)$ parameter draws, so the two bootstraps are resampled independently; this is conservative but does not preserve their covariance. The result remains conditional on the adopted electrical-capacitance upper bound, the static heating branch describing the driven device, and the electrical and $R(T)$ measurements belonging to the same sample. Relative to the $C = 0$ analysis, the upper-bound correction raises the central C_{th} estimate by only 1.5% (0.047163 to 0.047873 pJ/K), although the RMSE increases from 0.806 to 1.145 K. The 189.6 μA trace gives a smaller value and a substantially larger error because it approaches a reversal; including it would bias the single-heating-branch fit downward. The fitted 13.0 ns thermal time is of the same order as the measured 16–24 ns oscillation period. Yuanhang's thermal capacitance is approximately 1037 times larger.

The experimental manuscript reports a 150 nm film thickness and an approximately 200 nm electrode gap [2], but not the electrically active filament width or volume. A numerical geometry estimate $C_{\text{th}} = \rho c_p V$ would therefore introduce an unconstrained volume and is not reported.

7 Comparison across experimental currents

This is a blind prediction test: the fitted $R(T)$ curve, $T_0 = 314.4$ K, $S_e = 0.00367513$ mW/K, $C = 0.39$ pF, and $C_{\mathrm{th}} = 0.047873$ pJ/K are fixed before examining the result. Each baseline-corrected measured current waveform is replayed through the model; no parameter is adjusted by current. Predicted voltages are sampled at the experimental 1 ns times, and both data sets use the same 50–250 ns window and oscillation detector.

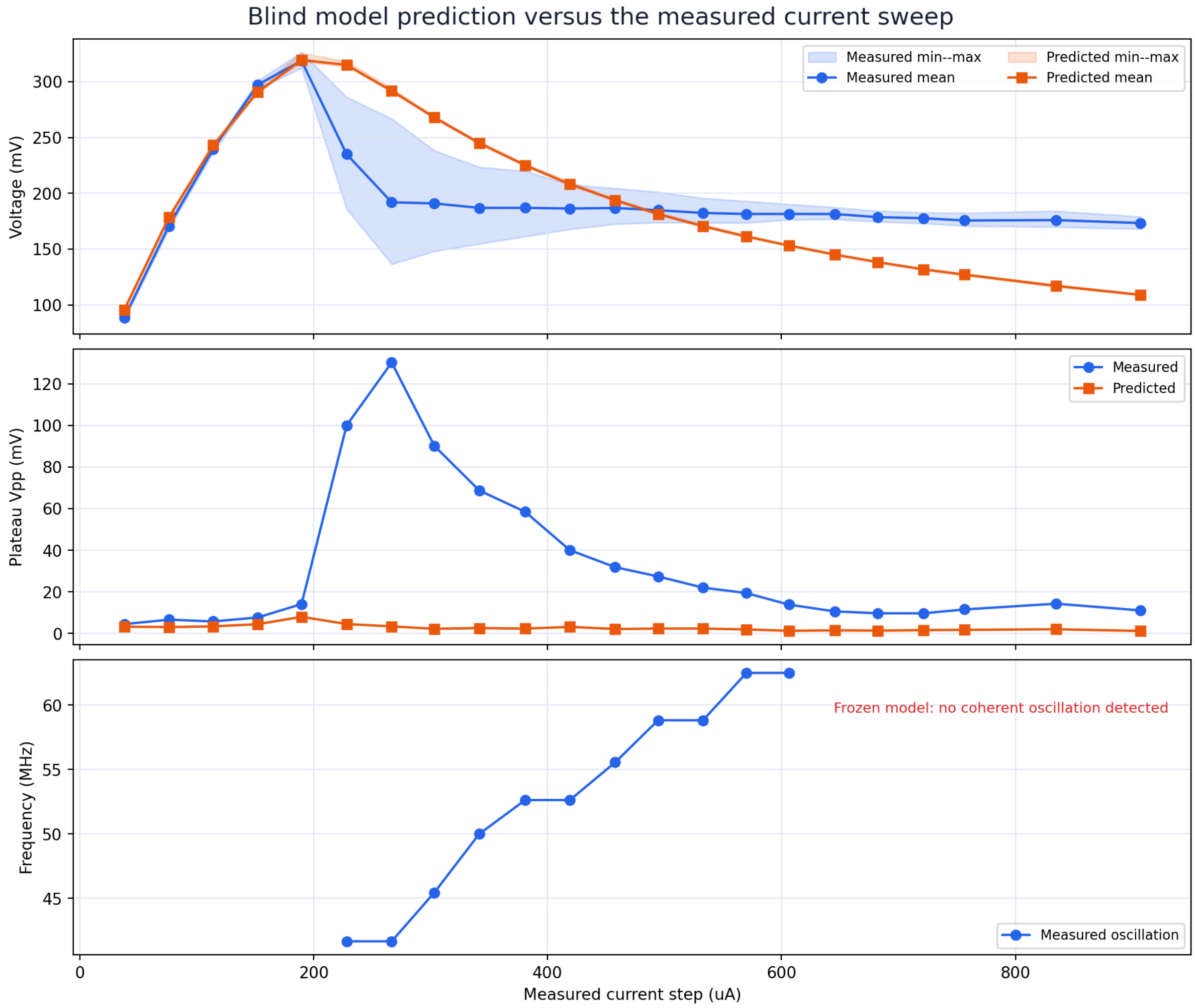

Figure 7: Measured behavior and blind model predictions over all 22 measured current steps. The shaded bands show the voltage extrema in the common analysis window.

The experiment oscillates over 11 records, from 228.2 to 606.3 μA, with frequencies of 41.7–62.5 MHz. The frozen model predicts no coherent oscillation at any measured current. It nevertheless reproduces the last stable pre-onset record: at 189.6 μA, the measured and predicted mean voltages are 318.86 and 319.53 mV. Thus the cold-side static calibration works, whereas the dynamic switching window does not.

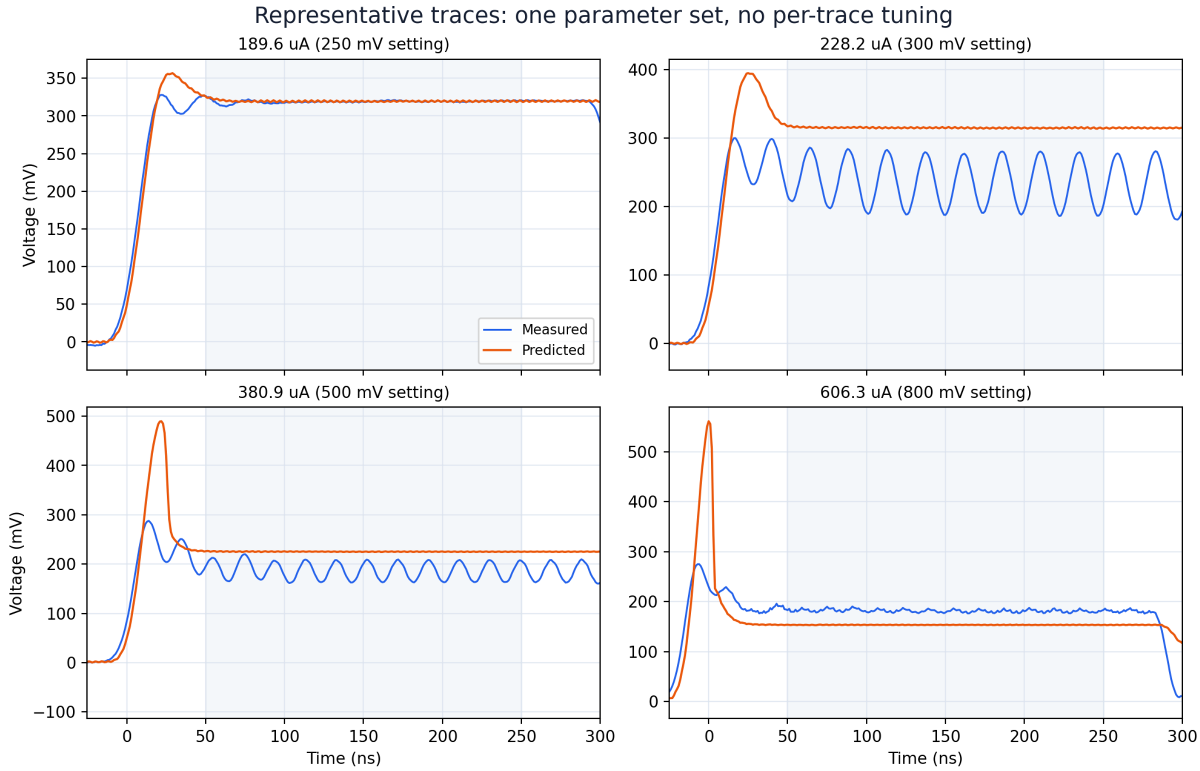

Figure 8: Representative measured and predicted voltages. The shaded interval is the common detection window. The 189.6 μA control is reproduced closely, but the measured oscillations above onset are replaced by steady model solutions.

Quantity	Experiment	Frozen model
Oscillating records	11 of 22	0 of 22
Oscillation-current interval	228.2–606.3 μA	none
Frequency interval	41.7–62.5 MHz	—
Energy per cycle in oscillating records	1.229–1.761 pJ	—
189.6 μA mean voltage	318.86 mV	319.53 mV
Median plateau-voltage RMSE	32.38 mV	
Maximum change after halving Δt	0.228 mV	

Table 8: Direct prediction results. Energy per cycle is the mean plateau power divided by the detected frequency.

Halving the integration step from 0.05 to 0.025 ns does not change any tested classification and changes the representative mean voltages by at most 0.228 mV. The missing oscillations are therefore not a time-step artifact.

8 Capacitance dependence and the thermal-only limit

In the zero-capacitance limit,

$$C = 0 \implies V(t) = I_{\text{in}}(t)R(T), \qquad \tau_{\text{th}} = \frac{C_{\text{th}}}{S_e}. \tag{16}$$

A thermal relaxation cycle requires the same current to heat through the heating transition and then cool through the cooling transition. Using the fitted branch resistances at the two transition centers gives

$$I > I_{\text{heat}} = \sqrt{\frac{S_e(T_{\text{heat}} - T_0)}{R_{\text{heat}}}} = 329.65\ \mu\text{A}, \qquad I < I_{\text{cool}} = \sqrt{\frac{S_e(T_{\text{cool}} - T_0)}{R_{\text{cool}}}} = 254.44\ \mu\text{A}. \quad (17)$$

Because $I_{\text{heat}} > I_{\text{cool}}$, these conditions do not overlap: the fitted system has no thermal-only oscillation window. At the adopted $C = 0.39$ pF, the metallic electrical time is $CR_m = 7.10$ ps, compared with $\tau_{\text{th}} = 13.026$ ns. The ratio is 5.45×10^{-4}, so this capacitance supplies negligible electrical memory and does not alter the steady thermal thresholds.

The numerical study varies C from 0 to 50 pF and C_{th} across its conditional 0.021918–0.092624 pJ/K interval while holding every other parameter fixed. The experimentally admissible range $0 \leq C \leq 0.39$ pF produces no oscillation anywhere in that interval. Oscillations first appear at 3 pF only at the lowest tested C_{th}; at the adopted C_{th}, they first appear at 7 pF. Both values exceed the electrical timing bound and would create a voltage lag absent from the measured edge.

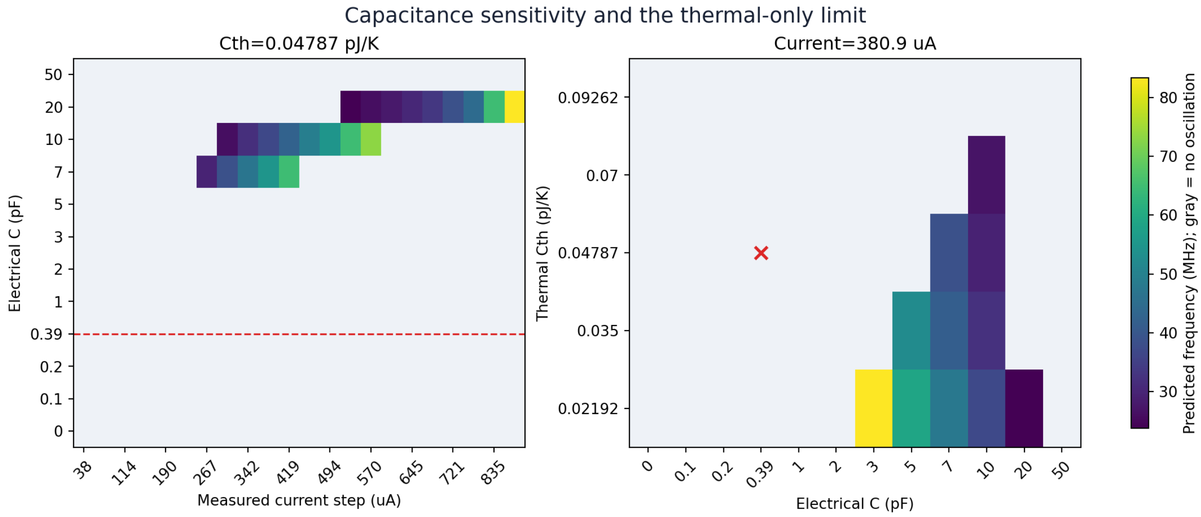

Figure 9: Predicted frequency under capacitance variation. Gray denotes no coherent oscillation. The dashed line and red cross identify the adopted electrical and thermal capacitances; larger electrical values are stress tests outside the timing bound.

The present parameter set therefore matches the stable pre-onset voltage but cannot reproduce the measured oscillatory regime without an electrical capacitance that contradicts the edge timing. The next useful tests are an independently measured dynamic switching loop and inclusion of the actual TIA/load impedance. Changing γ alone cannot explain the onset at 228.2 μA, because the predicted temperature there does not reach the first heating transition; γ acts only after a hysteresis reversal.

9 Global parameter inference from all measured waveforms

We next asked whether one shared set of model parameters could be inferred directly from the complete current–voltage data set. This is a stricter test than fitting one trace: the same vector

$$\boldsymbol{\theta} = (C, C_{\text{th}}, S_e, T_0, T_c, w, \beta, \gamma) \tag{18}$$

must describe all 22 measured current waveforms. The source settings at 200, 400, 600, 800, and 1000 mV were withheld from optimization, leaving 17 training traces and five blind test traces. The source-setting labels only identify records; the measured current waveform is the model input in every case.

For each candidate, the simulator uses the complete measured current transient and compares the predicted and measured voltage over the common 50–250 ns plateau. The weighted objective is

$$\begin{aligned} J = 0.20J_{\text{wave}} + 0.20J_{\text{mean}} + 0.15J_{\text{amp}} + 0.15J_{\text{spec}} \\ + 0.15J_f + 0.10J_{\text{class}} + 0.05J_{\text{edge}} + 0.05J_{\text{prior}}. \end{aligned} \tag{19}$$

Here J_{wave} is a normalized waveform error, with a ± 12 ns phase shift allowed only for oscillatory records; the other terms compare mean voltage, peak-to-peak amplitude, normalized spectrum, frequency, oscillation classification, and the 0–50 ns rising edge. The constrained fit includes a weak prior and keeps each parameter inside its independently estimated interval. A second, deliberately relaxed fit removes that prior and expands the bounds to test whether the model can fit the data only by abandoning those measurements. Differential evolution explores the global space, followed by bounded local refinement. Optimization uses $\Delta t = 0.2$ ns; every reported candidate is rerun at 0.05 ns.

Parameter	Starting value	Physical interval	Constrained fit	Relaxed fit
C (pF)	0.3900	0–0.390	0.2408	5.4187*
C_{th} (pJ/K)	0.04787	0.02192–0.09262	0.06146	0.10643*
S_e (mW/K)	0.003675	0.003434–0.004085	0.003621	0.004894*
T_0 (K)	314.400	314.25–314.55	314.416	317.844*
T_c (K)	333.494	333.137–333.838	333.476	332.927*
w (K)	6.8823	6.6291–7.2681	6.8726	2.5568*
β (K^{-1})	0.29916	0.28092–0.32021	0.29542	0.11613*
γ	0.95627	0.2–2.0	0.97956	1.03546

Table 9: Shared parameters inferred from all training traces. An asterisk marks a relaxed value outside the independently supported physical interval.

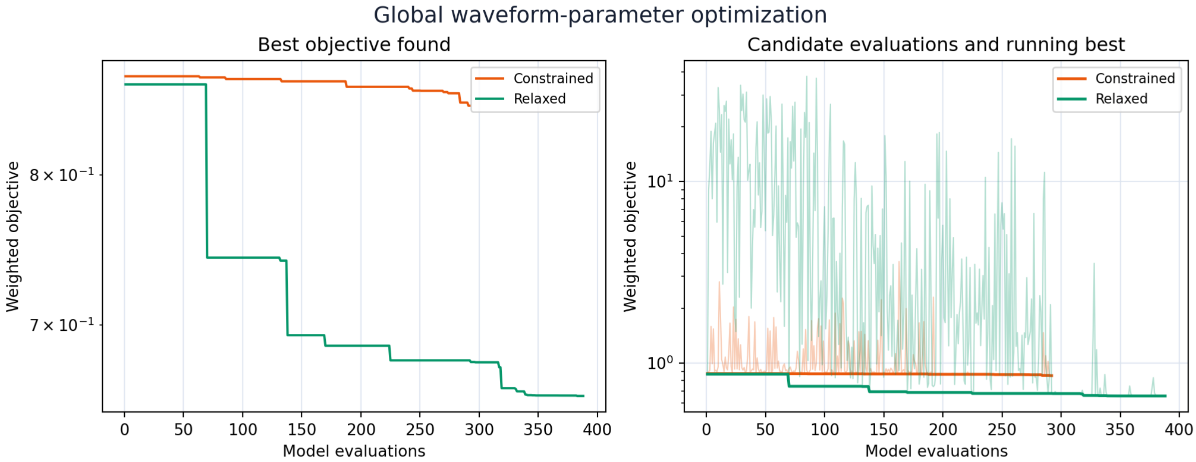

Figure 10: Optimization history for the physically constrained and relaxed searches. The large relaxed improvement occurs only after the search leaves the measured parameter region.

Model	Training J	Held-out J	All-trace J	Oscillations
Starting estimates	0.8680	0.9267	0.8813	0 of 22
Constrained fit	0.8601	0.9255	0.8750	0 of 22
Relaxed diagnostic	0.6515	0.7888	0.6827	0 of 22

Table 10: Fit and blind-test performance. Lower objective values are better. The experiment contains 11 oscillatory records.

The constrained fit lowers the all-trace objective by only 0.72% and the held-out objective by 0.13%. It remains essentially the independently estimated model and still predicts no oscillations. The relaxed search lowers the all-trace objective by 22.5% and the held-out objective by 14.9%, so its improvement is not confined to the training records. However, seven of its eight parameters lie outside their physical intervals, including $C = 5.42$ pF, and it also predicts no coherent oscillation. Its reduced amplitude error comes from large turn-on transients that settle to steady voltages.

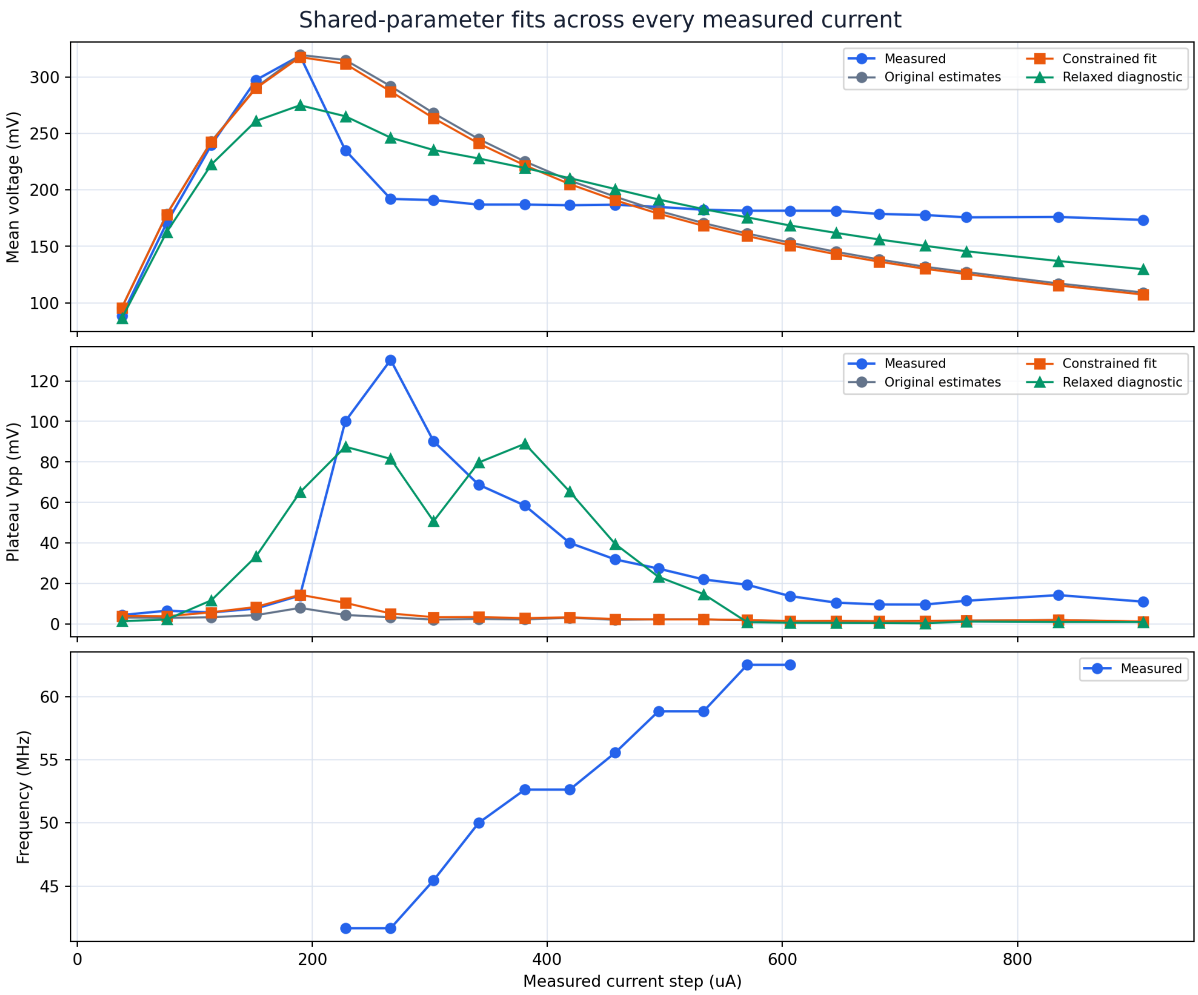

Figure 11: Measured and fitted operating summaries. The relaxed search partially reproduces mean voltage and apparent amplitude, but neither fit produces the measured frequency branch.

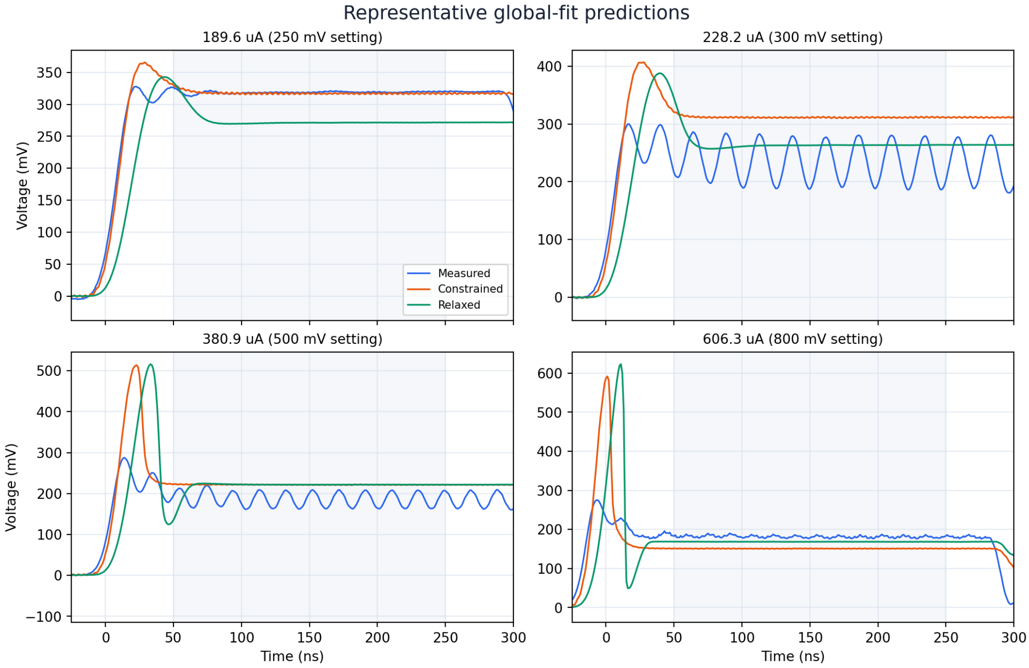

Figure 12: Representative waveform predictions. The relaxed fit can create a large initial excursion, but both fitted models settle instead of sustaining the measured oscillations.

Repeating the final evaluation at 0.2, 0.1, and 0.05 ns gives all-trace objectives of 0.6868, 0.6838, and 0.6827 for the relaxed fit, without changing any oscillation classification. The conclusion is therefore numerically converged. This exercise does not provide replacement physical parameter estimates: it shows that parameter adjustment within the present ideal-current, quasi-static-hysteresis model is insufficient. The likely missing information is dynamic switching behavior or the actual TIA/load impedance. Notably, γ remains near Yuanhang's value in both searches, so changing the minor-loop curvature alone is not the solution.

A Reproduction

The validation run is archived as

`20260817_100102_current-step-with-a-nonzero-metallic-voltage-val_6765e0`.

It can be reproduced from the repository root with

```
neuristor simulate current --config experiments/current/nonzero_voltage_valley.toml
```

The archived bundle contains the resolved configuration, traces, metrics, figure, and repository provenance. A fresh execution produced identical metrics and byte-identical trace data.

The resistance–temperature figure and synchronized GIF are regenerated from the same archived traces with `neuristor runs visualize`.

The sample resistance fit is archived as

`20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9`.

It is reproduced with

```
neuristor fit resistance --data data/experimental/100425_chip1_gap3.tsv --method
major-loop --bootstrap-samples 1000
```

The environmental-conductance analysis is archived as

`20260817_153807_environmental-thermal-conductance-estimate_761640`.

It is reproduced with

```
neuristor analyze conductance --data data/experimental/tia_current_sweep
--resistance-preset presets/resistance_100425_chip1_gap3.json
--resistance-bootstrap
public_jobs/20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9/parameter_bootstrap.csv
--ambient-K 314.4 --ambient-interval-K 314.25,314.55
```

The thermal-capacitance analysis is archived as

`20260828_112314_thermal-capacitance-estimate-with-conservative-0_aa2469`.

It is reproduced with

```
neuristor analyze thermal-capacitance --data data/experimental/tia_current_sweep
--resistance-preset presets/resistance_100425_chip1_gap3.json
--resistance-bootstrap
public_jobs/20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9/parameter_bootstrap.csv
--conductance-mW-per-K 0.003675126546984294
--conductance-bootstrap
public_jobs/20260817_153807_environmental-thermal-conductance-estimate_761640/conductance_bootstrap.csv
--ambient-K 314.4 --electrical-capacitance-pF 0.39
--selected-drives-mV 100,150,200 --fit-window-ns 15,35
```

The blind model comparison and capacitance study are archived as

`20260829_100718_specimen-model-prediction-versus-measured-curren_eefab7`.

They are reproduced together with

```
neuristor analyze model-validation --config
experiments/current/specimen_model_validation.toml
```

The global shared-parameter inference is archived as

`20260829_105704_global-specimen-parameter-inference-from-all-cur_8f12d6`.

It is reproduced with

```
neuristor analyze fit-waveforms --config
experiments/current/specimen_waveform_inference.toml
```

The bundle contains the resolved objective and bounds, every fitted parameter, per-trace metrics and predictions, held-out validation, optimization histories, figures, and the three-step convergence check.

The historical capacitance shortcut is preserved for provenance in

`20260817_134254_lab-current-trace-parameter-estimates_4223e0`.

Its `electrical_capacitance.csv` contains the four values reconstructed in Table 6. The bundle is retained as evidence of how 19.8 pF was obtained, but that estimate is superseded by the full current-balance analysis in Section 5.

References

[1] Y.-H. Zhang, C. Sipling, E. Qiu, I. K. Schuller, and M. Di Ventra, "Collective dynamics and long-range order in thermal neuristor networks," *Nature Communications*, vol. 15, art. 6986, 2024. doi:10.1038/s41467-024-51254-4.

[2] A. Gildor, S. Hodisan, S. Kvatinsky, and Y. Kalcheim, "Harnessing the VO_2 Phase Transition for Automatic Gain Control in Transimpedance Amplifiers," manuscript, arXiv:2604.04594v1, 2026.